AI 101 for Mechanical Engineers · Module 5

Features and repeatable pipelines

Lesson 5.3

What you will learn

> Construct meaningful engineering features, enforce their availability at prediction time, and use a repeatable pipeline with documented input requirements.

Lesson overview

Module 5 · Lesson 3

Features and repeatable pipelines

AI 101 for Mechanical Engineers

1 Build features from engineering knowledge

Create inputs from physical relationships and available measurements.

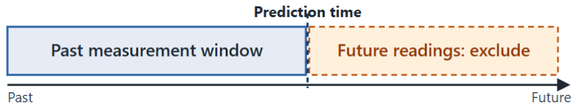
$$P_h = \Delta p \times Q$$

Δp: pressure rise (Pa)
Q: volumetric flow rate (m³/s)
P_h: hydraulic power (W)

Pressure rise | Hydraulic power | Vibration RMS | Frequency-band energy | Temperature above ambient

Check that each feature adds value for the prediction task.

2 Use only available measurements



Record the window bounds and prediction time.

3 Split, fit, then apply

1. **Split the data**
   Keep training, validation and test examples separate.
2. **Fit on training folds only**
   Learn scaling, imputation and feature-selection parameters.
3. **Store the fitted rules**
   Keep the learned parameters.
4. **Apply without refitting**
   Use the stored rules on validation and test inputs.

4 Record the input contract

State what the stored model expects:

Expected units	The unit for each input feature
Missing-value rules	How missing inputs are handled
Modified values	Which values changed, and the rule used
Derived features	The engineering reason for each one

Example: if power changes from W to kW, apply the expected conversion.

Caution: a pipeline does not fix an invalid split or leaked information.

Key takeaway — **Use meaningful features, respect prediction time and reuse training-only transformations.**

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Build features from engineering knowledge

- Constructs useful inputs using problem knowledge: “hand-crafted feature extraction” [10]
- Examples: pressure rise, hydraulic power, vibration RMS
- Also: frequency-band energy, temperature above ambient
- Must be meaningful and available at prediction time

Hydraulic power combines measured quantities physically

$$P_h = \Delta p \times Q$$

- Δp: pressure rise in Pa
- Q: volumetric flow rate in m^3/s
- P_h: hydraulic power in W

Generate feature ideas systematically

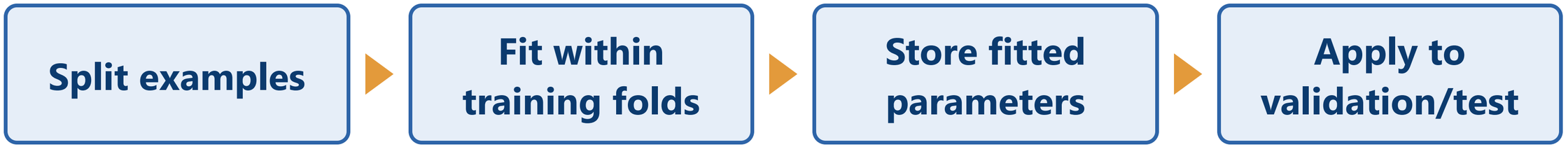
1. Start from the decision and relevant physical relationships
2. Consider products or ratios with clear units and assumptions
3. Use changes over time if available at decision time
4. Use window statistics when signal history is relevant
5. Evaluate added value; do not include features just because calculable

Prediction windows must exclude future samples

- A windowed feature uses a specified time interval
- For an earlier alarm, the interval excludes future samples
- Later measurements leak information unavailable during use
- Document the window and the decision time

> A physically plausible feature can still be invalid if information arrives too late.

Split, fit, store, then apply



- A pipeline bundles preprocessing and modelling repeatably [34]
- Fit scalers, imputers and learned feature selection on training folds
- Refitting a scaler on final-test data crosses the boundary, even without targets

A pipeline cannot fix an invalid split

- A scikit-learn Pipeline keeps preprocessing and estimator together
- Fitting and prediction then use the same steps
- It must still be fitted inside each training portion
- Wrapping a leaky workflow does not fix it

Record the input contract

- States what the stored model expects, including units
- Includes the missing-value rules
- A change from W to kW may be misread without conversion
- Keep units, modified values and feature justifications traceable

CHECK YOUR UNDERSTANDING

Which statistics should a scaler use when transforming the final test set?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Which statistics should a scaler use when transforming the final test set?

> Use the stored statistics fitted on training data. Apply them unchanged rather than fitting the scaler on the final test set.

CHECK YOUR UNDERSTANDING

Can a signal feature include measurements taken after the intended alarm time?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Can a signal feature include measurements taken after the intended alarm time?

> No. That would introduce future information unavailable when the alarm had to be issued.

Key takeaways

- Construct meaningful features using available engineering information
- Exclude future measurements from earlier prediction windows
- Separate training-only fitting from later application
- Document expected units, missing-value rules and modified values

References

[10] Y. Lei, B. Yang, X. Jiang, F. Jia, N. Li, and A. K. Nandi, “Applications of machine learning to machine fault diagnosis: A review and roadmap,” Mech. Syst. Signal Process., vol. 138, Art. no. 106587, 2020. doi: 10.1016/j.ymssp.2019.106587

[34] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” J. Mach. Learn. Res., vol. 12, pp. 2825–2830, 2011.

Numbers match the course reference list. The written lesson is the authoritative text.